[Model] Unified Finsler Triplet Loss Formulation

Finsler distance, RandomIdentitySampler, batch-hard mining on pid

$$\mathbf{x}_i = (\mathbf{x}_i^{\mathrm{id}}\ \mathbf{x}_i^{\omega})^{\intercal}$$

$$\mathcal{L}_{tri}^{uni} = \frac{1}{B}\sum_{i}^{N}\left[\max_{p:y_p=y_i} d_F(\mathbf{x}_i,\mathbf{x}_p) - \min_{n:y_n\neq y_i} d_F(\mathbf{x}_i,\mathbf{x}_n) + m_{id}\right]_+$$

$$d_F(\mathbf{x},\mathbf{y}) = \|\mathbf{x}_{\mathrm{id}} - \mathbf{y}_{\mathrm{id}}\|_2 + \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id}\rangle$$

"We want to compare images from different domains, not necessarily same identity"

- Loss can be formulated as a cross-domain constrastive on identity features via Finsler distance
- Encourages the Finsler distance to shrink cross-domain same-identity pairs
    - Asymmetric term must compensate for the domain drift

[Model] Modified Triplet Loss Terms

Motivation: Clean signal separation between identity-related and domain-related loss terms

Idea 1: Split Triplet Loss into Identity-Euclidean + Domain Finsler

$$\mathbf{x}_i = (\mathbf{x}_i^{\text{id}}\ \mathbf{x}_i^{\omega})^{\intercal}$$

1a. Euclidean Identity Triplet: Euclidean distance, RandomIdentitySampler, batch-hard mining on pid

$$\mathcal{L}_{tri}^{id} = \frac{1}{B}\sum_{i}^{N}\left[\max_{p:y_p=y_i} d_E(\mathbf{x}_i^{\text{id}}, \mathbf{x}_p^{\text{id}}) - \min_{n:y_n\neq y_i} d_E(\mathbf{x}_i^{\text{id}}, \mathbf{x}_n^{\text{id}}) + m_{id}\right]_+$$

1b. Finsler Domain Triplet: Finsler distance, RandomDomainSampler, batch-hard mining on domain/camera

$$\mathcal{L}_{tri}^{dom} = \frac{1}{B}\sum_{i}^{B}\left[\max_{p:d_p=d_i} d_F(\mathbf{x}_i, \mathbf{x}_j) - \min_{n:d_n\neq d_i} d_F(\mathbf{x}_i, \mathbf{x}_n) + m_{dom}\right]_+$$

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\text{id}} - \mathbf{y}_{\text{id}}\|_2 + \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id}\rangle$$

d_i Is the domain/camera label

- Facilitates feature disentanglement and isolates the contributions of asymmetric (Finsler) geometry to domain modelling

Rationale
- Identity invariance stays Euclidean (same-person features close regardless of domain)
- Domain structure is modeled via Finsler (same-domain features clustered in asymmetric space)

Consideration
- We can still fine-tune the margin for Finsler Domain Triplet

Critical analysis
- Domain triplet pulls same-domain embeddings closer and pushes different-domain embeddings apart
- Existing domain loss does the exact opposite; it repels nearest same-domain memory bank neighbors
- If we proceed with domain triplet we will be clustering by domain or camera, which is the failure mode of domain-generalizable person re-identification.
    - Re-introduce camera-style shortcuts
    - Loss learns to cluster arbitrary identities by domain

If we want to learn what is different between domains (drift vector models domain artifacts)
- Positive should be cross-domain same-identity
- Negative should be same-domain same-identity

[Model] Modified Triplet Loss Terms

Motivation: Clean signal separation between identity-related and domain-related loss terms

Idea 1: Split Triplet Loss into Identity-Euclidean + Domain Finsler

$$\mathbf{x}_i = (\mathbf{x}_i^{\mathrm{id}}\ \mathbf{x}_i^{\omega})^{\intercal}$$

1a. Euclidean Identity Triplet: Euclidean distance, RandomIdentitySampler, batch-hard mining on pid

$$\mathcal{L}_{tri}^{id} = \frac{1}{B}\sum_{i}^{N}\left[\max_{p:y_p=y_i} d_E(\mathbf{x}_i^{\mathrm{id}}, \mathbf{x}_p^{\mathrm{id}}) - \min_{n:y_n\neq y_i} d_E(\mathbf{x}_i^{\mathrm{id}}, \mathbf{x}_n^{\mathrm{id}}) + m_{id}\right]_+$$

1b. (Refined) Cross-domain constrastive loss: Finsler distance, RandomDomainSampler

$$\mathcal{L}_{contrast}^{dom} = \frac{1}{|\mathcal{C}|}\sum_{(i,j)\in\mathcal{C}} d_F(\mathbf{x}_i, \mathbf{x}_j)^2 \qquad \mathcal{C} = \{(i,j) : y_i = y_j, d_i \neq d_j\}$$

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\mathrm{id}} - \mathbf{y}_{\mathrm{id}}\|_2 + \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id}\rangle$$

d_i Is the domain/camera label

"We want to compare images from different domains, not necessarily same identity"

- Loss can be formulated as a cross-domain constrastive on identity features via Finsler distance
- Encourages the Finsler distance to shrink cross-domain same-identity pairs
    - Asymmetric term must compensate for the domain drift

[Model] Modified Triplet Loss Terms

Motivation: Clean signal separation between identity-related and domain-related loss terms

Idea 1: Split Triplet Loss into Identity-Euclidean + Domain Finsler

$$\mathbf{x}_i = (\mathbf{x}_i^{\mathrm{id}}\ \mathbf{x}_i^{\omega})^{\intercal}$$

1a. Euclidean Identity Triplet: Euclidean distance, RandomIdentitySampler, batch-hard mining on pid

$$\mathcal{L}_{tri}^{id} = \frac{1}{B}\sum_{i}^{N}\left[\max_{p:y_p=y_i} d_E(\mathbf{x}_i^{\mathrm{id}}, \mathbf{x}_p^{\mathrm{id}}) - \min_{n:y_n\neq y_i} d_E(\mathbf{x}_i^{\mathrm{id}}, \mathbf{x}_n^{\mathrm{id}}) + m_{id}\right]_+$$

1c. Euclidean L2 Loss on Drift Vector: Euclidean distance, RandomDomainSampler

$$\mathcal{L}_{\text{drift-xcontrast}} = \frac{1}{|\mathcal{C}|}\sum_{(i,j)\in\mathcal{C}} \|\boldsymbol{\omega}_i - \boldsymbol{\omega}_j\|_2^2 \quad \mathcal{C} = \{(i,j) : y_i = y_j,\ c_i \neq c_j\}$$

"We want to compare images from different domains, not necessarily same identity"

- Loss can be formulated as a cross-domain constrastive on identity features via Finsler distance
- Encourages the Finsler distance to shrink cross-domain same-identity pairs
    - Asymmetric term must compensate for the domain drift

[Model] Modified Triplet Loss Terms -- Results

Short Name	Description	Drift Conditioning	mAP	Rank-1
baseline_1a_only	Euclidean identity triplet only (Idea 1a); no cross-domain contrastive; no domain triplet; no drift-only L2	instance	43.7	43.4
		domain	44.1	45
cam_crossview_contrast	Refined 1b: Finsler distance on same-PID pairs with different global camera id; weight 0.1	instance	42.2	42.2
		domain	42.8	44.1
domain_triplet_no_memory_domain	Batch-hard triplet on camera labels using full Finsler embedding; domain_triplet_weight=1.0; BAU memory domain loss disabled (--no-domain)	instance	24.6	30.4
		domain	24.6	30.4
drift_only_cross_camera_l2	Mean squared L2 on drift vectors for same PID different camera; weight 0.1	instance	43	43.1
		domain	42.8	41.9
domain_triplet_with_memory_domain	Same domain triplet as 3rd but BAU memory domain loss enabled (no --no-domain)	instance	35.5	36.4
		domain	36.2	36.1

[Model] Finslerized ReNorm

short_name	drift_conditioning	mAP	Rank1	eval_epoch
Baseline euclidean	Final Epoch	42.36	41.5	119
	Peak MAP	43.05	42.71	79
unified_finsler	instance			
	domain	42.21	41.79	69
1a + 1b	instance			
	domain	35.55	34.29	69
1a + 1b (refined)	instance	14.76	13.29	59
	domain	32.01	32.14	39
	domain	28.81	29.29	59

This Week

1. [Results] Finslerized ReNorm – Results
2. [Results] Modified Triplet Loss with Euclidean Domain Loss
3. [Results] Asymmetric Ranking during Evaluation
4. [Results] Best model configuration vs. Protocol-2
5. [Theory] Intuition/Justification on why this config improves on Euclidean BAU

Experiments if we want to publish

1. [Sweep] Softmax temperature and domain token loss weight
2. [Ablation] Instance Normalization Layers
3. [Sweep] Omega regularizer scalar weight
4. [Sweep] Scalar multiplier for residual correction term

[Model] Finslerized ReNorm

Name	Drift Conditioning	mAP		Rank1	
Baseline Euclidean	N/A	42.36		41.5	
Unified finsler	instance	40.85	-1.51	40.43	-1.07
	domain	40.84	-1.52	40.50	-1.00
1a + 1b	instance	33.02	-9.34	32.93	-8.57
	domain	33.39	-8.97	33.00	-8.50
1a + 1b (refined)	instance	17.12	-25.24	16.64	-24.86
	domain	26.23	-16.13	26.14	-15.36

- The core Finsler components (split identity/drift head, symmetric_trapezoidal distance, batch-hard mining on asymmetric distance matrix) were ported into ReNorm2 to test the Finsler hypothesis in a different training environment
- Motivation: BAU's uniformity loss was identified as actively suppressing drift. ReNorm has no uniformity loss, so it provides a cleaner test of whether asymmetry survives when the training environment is not hostile to it

Two high-severity confounds were documented before any experiments ran:

- Finsler triplet uses L2-normalized identity features; the Euclidean baseline uses unnormalized — this creates a different optimization landscape independent of the distance geometry
- ReNorm's RN+EN feature fusion at eval produces non-unit-normed identity vectors, which breaks the Finsler distance's assumed scale relationship between base and asymmetry term

[Model] Modified Triplet Loss Terms

Short Name	Description	Drift Cond.	Fins. Dom.		Euc. Dom.	
			mAP	R1	mAP	R1
1a	Baseline: Euclidean identity triplet only.	instance	**44.1**	43.6	**43.9**	**43.9**
		domain	43.2	43.7	43.8	43.9
1b	Domain Triplet Only: Finsler triplet mining camera labels	instance	35.1	33.9	30.6	30.6
		domain	33.2	34.1	25.6	27.0
1a + 1b	Conflicting Objectives: Both identity and camera triplets are active simultaneously.	instance	36.4	36.2	36.9	36.2
		domain	34.9	36.1	37.8	38.9
1a + 1b (no L_dom)	Same as above, but disabling BAU domain repulsion to isolate domain-attraction.	instance	34.6	36.4	33.6	35.0
		domain	24.9	28.5	25.5	26.4
1a + Refined 1b	Cross-Camera Contrastive: Shrinks Finsler distance d_F for the *same* identity across *different* cameras	instance	43.1	43.7	42.8	43.4
		domain	42.1	43.2	42.4	43.1
1a + 1c	Drift L_2 Alignment: Explicitly penalizes the L_2 variance of drift vectors for the same identity across cameras.	instance	43.0	43.4	43.8	42.9
		domain	43.8	**44.9**	43.7	42.9
Unified Finsler	Single Finsler Triplet: Replaces the Euclidean identity metric with the full asymmetric Finsler distance d_F.	instance	**44.1**	44.0	43.4	43.9
		domain	**44.1**	44.3	43.4	43.0

Instance Conditioning: RandomIdentitySampler; Domain Conditioning: RandomMultipleGallery

- Orange box shows similar values, which is expected because in both runs, the Finsler injection to BAU is removed such that these runs should be identical.
    - Resnet50_finsler acts as a structurally larger model -> uses a split-head architecture; even when the drift is not explicitly evaluated, the backbone is producing a feature vector of 4096 dimensions
    - Orthgonalization step ensures that the drift branch remains orthogonal to the identity branch; strong regularizer on the convolutional backbone -> forces it to learn a more robust representation
    - Normalization performed *after* slicing the identity dimension -> magnitude distribution before normalization differs

- Green boxes should be identical, since the domain loss is deactivated
    - mAP varies significantly -> arguably due to the conflicting retrieval objectives making the optimization unstable, distinct local minimas?

[Model] Modified Triplet Loss Terms - Analysis

1. **Explicitly Camera Clustering (1b only)**: Pulling same-camera features together forces model to retain domain-specific variation, even in the identity slice.
   - Decrease: Rank-1 (-9.65|-15.10) and mAP (-9.5|-15.75) compared to 1a
2. **Gradient Conflict (1a+1b)**: For any pair (x_i, x_j) where the identity is same but camera is different
   - $\mathcal{L}_{1a}$ treats them as a positive pair (gradient pulls them together); $\mathcal{L}_{1b}$ treats them as a negative pair (gradient pushes them apart).
   - Decrease: Rank-1 (-8.0|-6.35) and mAP (-7.5|-6.5) compared to 1a
3. **Gradient Conflict (1a+1b w/ Ldom)**: Domain loss uniformly distributes (locally) the features of each dataset, whereas 1b promotes camera-based clustering (subset of dataset label)
   - Increases: Rank-1 (+5.9|+7.3) and mAP (+3.7|+7.35) compared to w/o Ldom
4. **Best Performer (1a + 1c)**: Yields observed only in domain conditioning + Finsler domain loss
   - Increases: Rank-1 (+0.6) and mAP (+1.2) compared to 1a
5. **Unified Finsler :** With Finslerized Domain Loss, Unified Finsler formulation outperforms/matches the baseline. With Euclidean Domain Loss, it falls matches/falls shortly behind (~0.5)
6. **Finsler vs. Euclidean Domain Repulsion**: Exact contribution of Finsler Domain Repulsion unclear vs. RNG

Naïve Conclusion
When domain-conditioning the drift vectors, they should undergo camera-based repulsion rather than clustering for generalizability, but identity-based clustering (1, 2, 3, 4)

1. Decrease indicates a collapse in generalizability
   - *Hypothesis:* Destroys the identity manifold into domain-specific sub-manifolds, directly violating the primary objective of extracting *domain-invariant* identity features.
2. 1a promotes identity-based clustering, while 1b promotes camera-based clustering
   - *Hypothesis*: Gradient interference stalls the network
   - Destructive interference (instead of regularizing) -> theoretically justifies the orthogonal decoupling of identity and drift vector slice in the drift head
3. Gradient interference helps regularize the network
   - Disabling L_dom causes a performance collapse when using domain conditioning, regardless of whether the metric space is Euclidean or Finslerized.
   - When using instance conditioning, removing L_dom produces marginal shifts,
     - **Conclusion**: Domain-conditioned drift fundamentally requires BAU domain

repulsion (L_dom) to prevent feature collapse, whereas instance-conditioned drift can survive without it; presumably as regularizer to prevent degeneration of the drift head

4. (1a+1c): 1a + 1c with domain drift conditioning achieves the highest Rank-1 in the entire sweep: 44.9 under Finsler-domain training, 45.0 under Finsler eval
    - light drift regularization that keeps the drift space free but directionally coherent is better than either no constraint (pure 1a) or hard disentanglement (1b)

5. It does so without requiring an explicit Euclidean Identity Triplet slice.
    - *Hypothesis*: The asymmetric Finsler distance, with a regularized multiplier omega, naturally resolves identity and domain slice distinction and domain-shift magnitude/direction on its own, rendering rigid subspace partitioning redundant.

[Model] Asymmetric Ranking during Evaluation - Theory

If `--eval-drift true` is going to improve performance, the drift vector must have received meaningful gradient signals to encode non-commutative domain shifts without collapsing to zero, and the core identity features' invariance must not be destroyed.

Two evaluation modes for resnet50_finsler
--eval-drift false (default during training):
- The model returns only the **identity slice** (2048-d, L2-normalized)
- Ranking uses standard **Euclidean distance** on identity features only
- Drift head is trained but invisible at retrieval time

--eval-drift true (post-hoc re-evaluation):
- The model returns the **full embedding** [identity | drift] (4096-d)
- Ranking uses finsler_drift_dist — Euclidean base + asymmetric drift term
- Probe→gallery and gallery→probe distances are now different

- The select_eval_embedding() function in the model is the switch: it either returns the full [identity_norm | drift] concatenation or slices off just the identity portion, depending on the use_drift_in_eval flag set by --eval-drift
- Training always uses both slices for the triplet and domain losses; this flag only controls what the evaluator sees
- The sweep was run as a post-hoc batch job (sweep_eval_drift_true_consecutive.sbatch) on all saved best.pth checkpoints from the Idea-1 sweeps — it did not retrain anything, just re-evaluated
- The near-zero ΔmAP result is the key diagnostic: the drift vectors are being trained but are not encoding asymmetric information that changes retrieval rankings

[Model] Asymmetric Ranking during Evaluation Results (w/ Finsler Domain Loss)

		Fins. Dom.					
		mAP			Rank-1		
Short Name	Best cond.	Eucl. Eval	Fins. Eval	Δ	Eucl. Eval	Fins. Eval	Δ
1a	instance	44.1	44.1	+0.0	43.6	43.6	+0.0
1b	instance	35.1	35.1	+0.0	33.9	33.9	+0.0
1a + 1b	instance	36.4	36.5	+0.1	36.2	36.3	+0.1
1a + 1b (no L_dom)	instance	34.6	34.6	+0.0	36.4	36.6	+0.2
1a + Refined 1b	instance	43.1	43.2	+0.1	43.7	43.6	−0.1
1a + 1c	domain	43.8	43.8	+0.0	44.9	45.0	+0.1
Unified Finsler	instance	44.1	44.1	+0.0	44.0	44.1	+0.1

[Model] Asymmetric Ranking during Evaluation Results (w/ Euclidean Domain Loss)

		Eucl. Dom					
		mAP			Rank-1		
Short Name	Best cond.	Eucl. Eval	Fins. Eval	Δ	Eucl. Eval	Fins. Eval	Δ
1a	Instance	43.9	43.9	+0.0	43.9	43.9	+0.0
1b	Instance	**30.6**	**24.5**	**−6.1**	**30.6**	**27.7**	**−2.9**
1a + 1b	Domain	37.8	37.8	+0.0	38.9	39.1	+0.2
1a + 1b (no L_dom)	Instance	33.6	33.6	+0.0	35.0	34.7	−0.3
1a + Refined 1b	Instance	42.8	42.8	+0.0	43.4	43.4	+0.0
1a + 1c	Instance	43.8	43.8	+0.0	42.9	42.9	+0.0
Unified Finsler	instance	43.4	43.4	+0.0	43.9	43.5	−0.4

- The single clearest pattern across all 28 experimental arms: switching from Euclidean to Finsler ranking at test time (ΔmAP) is essentially zero everywhere, except for the 1b arm under Euclidean-domain training where it drops by −6.1 mAP — revealing that when the domain loss suppresses the drift vector during training, activating Finsler ranking at test time actively harms performance

[Model] Asymmetric Ranking during Evaluation – Full Results

		Fins. Dom.				Euc. Dom.			
		Euclidean Eval		Finsler Eval		Euclidean Eval		Finsler Eval	
Short Name	Drift Cond.	mAP	R1	mAP	R1	mAP	R1	mAP	R1
1a	instance	**44.1**	43.6	**44.1**	43.6	**43.9**	**43.9**	**43.9**	**43.9**
	domain	43.2	43.7	43.3	44.0	43.8	43.9	43.8	43.9
1b	instance	35.1	33.9	35.1	33.9	30.6	30.6	24.5	27.7
	domain	33.2	34.1	33.3	34.2	25.6	27.0	21.6	24.5
1a + 1b	instance	36.4	36.2	36.5	36.3	36.9	36.2	36.9	36.3
	domain	34.9	36.1	34.8	35.6	37.8	38.9	37.8	39.1
1a + 1b (no L_dom)	instance	34.6	36.4	34.6	36.6	33.6	35.0	33.6	34.7
	domain	24.9	28.5	24.9	28.4	25.5	26.4	25.4	26.1
1a + Refined 1b	instance	43.1	43.7	43.2	43.6	42.8	43.4	42.8	43.4
	domain	42.1	43.2	42.1	43.0	42.4	43.1	42.3	42.9
1a + 1c	instance	43.0	43.4	43.0	43.4	43.8	42.9	43.8	42.9
	domain	43.8	**44.9**	43.8	**45.0**	43.7	42.9	43.7	42.9
Unified Finsler	instance	**44.1**	44.0	44.1	44.1	43.4	43.9	43.4	43.5
	domain	**44.1**	44.3	44.1	44.2	43.4	43.0	43.1	42.2

[Results] Best Model Configuration vs. Protocol-2

	M+MS+CS → C3		M+CS+C3 → MS		MS+CS+C3 → M		Average	
	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
CLIP-FGDI	**44.4**	44.6	**31.1**	**59.4**	**79.4**	**91.3**	**51.6**	**65.1**
ReNorm	43.6	44.7	25.6	55.6	72.7	89.1	47.3	63.1
ReNorm*	42.4	41.5	24.2	53.3	72.6	88.9	46.4	61.2
BAU	42.8	43.9	24.3	50.9	77.1	90.4	48.1	61.7
BAU*	42.21	42.5	23.28	49.70	76.61	89.61	47.37	60.60
Finsler BAU[1]	43.95	43.93	22.22	48.04	76.47	89.91	47.55	60.63
Finsler BAU[2]	44.27	44.57	22.79	49.16	76.26	89.28	47.77	61.00

- resnet50_finsler, domain drift conditioning, RandomMultipleGallery sampler, omega regularization (weight 1.5), symmetric_trapezoidal drift method, identity-triplet-only loss, bidirectional triplet, Finsler memory bank (--memory-bank-mode full), **eval-drift false**

Main Takeaways

1. The post-hoc eval-drift sweep confirmed that the trained drift vector is **not contributing to ranking**: switching to Finsler eval changes mAP by ≤ 0.1— the asymmetric distance component is not yet doing useful work at retrieval time
   - Increase in performance may be attributed to increase in model capacity
2. The 1b (domain triplet) failure is robust and generalizes across both domain loss types; explicit clustering by camera during training reliably breaks generalization
3. Best performing run this week: **1a + 1c with domain conditioning** — small but consistent Rank-1 improvement via light drift regularization, no mAP penalty
4. The previous slide table had systematic mAP/R1 column swaps in 20/56 cells; the table is now verified against raw logs, all 112 values confirmed correct
5. The ReNorm2 port provides a candidate cross-framework test, but two high-severity confounds mean its results cannot yet be cleanly interpreted — addressing those (normalizing identity at eval, fixing training normalization parity) is a prerequisite

- resnet50_finsler, domain drift conditioning, RandomMultipleGallery sampler, omega regularization (weight 1.5), symmetric_trapezoidal drift method, identity-triplet-only loss, bidirectional triplet, Finsler memory bank (--memory-bank-mode full), **eval-drift false**

[Report] Overview and Questions

1. Initial draft on Overleaf – when should I share access?
2. Triplet loss formulation does not fit onto a single line, is splitting into two lines acceptable?

3.1. Euclidean triplet baseline

Standard ReID pipelines construct a symmetric metric space via batch-hard triplet loss [6]; we retain this formulation as the controlled reference for ablation. Let $f_\theta : \mathcal{X} \to \mathbb{R}^d$ map an image $\mathbf{x}_i$ to an L_2-normalized embedding $\mathbf{z}_i$:

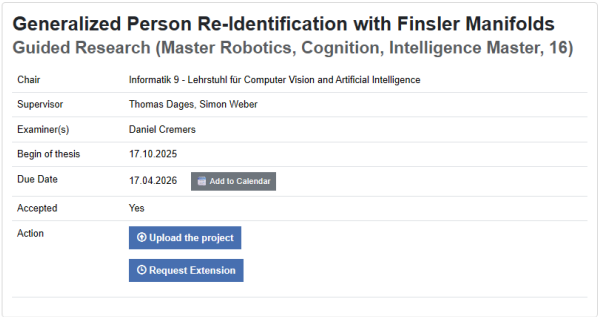
$$\mathcal{L}_{tri} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left[\max_{p: y_p = y_i} d_E(\mathbf{z}_i, \mathbf{z}_p) - \min_{n: y_n \neq y_i} d_E(\mathbf{z}_i, \mathbf{z}_n) + m \right]_+ , \quad (3)$$

where $[\cdot]_+ = \max(\cdot, 0)$, $d_E(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_2$, $\mathcal{B}$ is the mini-batch, and $m > 0$ is the margin.

[Administrative] Requirements of the report

1. **Length**: 8-12 pages in English
2. **Submission deadline**: Max. 6 months (17.04.2026), directly to the supervisor

Is there a presentation?



Generalized Person Re-Identification with Finsler Manifolds

Guided Research (Master Robotics, Cognition, Intelligence Master, 16)

Chair	Informatik 9 - Lehrstuhl für Computer Vision and Artificial Intelligence
Supervisor	Thomas Dages, Simon Weber
Examiner(s)	Daniel Cremers
Begin of thesis	17.10.2025
Due Date	17.04.2026 Add to Calendar
Accepted	Yes
Action	Upload the project Request Extension

Next Week

1. Retry 1b but detach the identity slice such that backpropagation on this loss term only influences the drift vector
2. Read up on feature disentanglement in this particular domain – what else am I missing?
3. Continue writing the report

Prove: that Finsler asymmetry captures domain drift without corrupting ID invariance,

1. **Toy Dataset Construction (Sanity Check):** Generate a synthetic dataset with controlled, non-commutative corruptions (e.g., progressive Gaussian blur, illumination down-scaling) applied to identical PIDs.
2. **Disentanglement Validation:** Extract embeddings from the Toy Dataset. Compute the variance of the 2048-d ID vector and the N-d drift vector. The ID vector variance must approach zero across corruptions, while the drift vector magnitude must correlate monotonically with corruption severity.
3. **Asymmetric Evaluation Activation:** Execute parallel sweeps on Protocol-2 utilizing both --eval-drift false and --eval-drift true. Quantify the delta in mAP. If Finsler asymmetry functions correctly, --eval-drift true will yield superior performance on unseen target domains by explicitly accounting for the source-to-target domain shift during the ranking phase.

1. When configuring the evaluation with --eval-drift true, how does the specific Finsler distance function utilized in this repository mathematically compute the asymmetric distance between a source query and an unseen target gallery?
2. For the planned Toy Dataset, what specific synthetic corruptions are you selecting to model non-commutative visual domain gaps that strictly necessitate an asymmetric distance metric over a symmetric one?
3. In the absence of cross-dataset identities in Protocol-2, how are you ensuring that the drift vector learns a generalized representation of "domain shift" rather than simply memorizing the specific camera parameters of the source training sets?